

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
CHANGA

First Semester of M.Tech. (CSE). (Backlog) Examination
May 2011

EC – 704 Digital Signal Processing and Applications

Date: 12.05.11, Thursday

Time: 10.00 a.m to 1.00 p.m

Maximum Marks: 70

Instructions: (1) Attempt all questions.

(2) Answer to the two sections must be written in separate answer books.

(3) Figures to the right indicate full marks

(4) Make suitable assumptions and draw neat figures whenever required.

SECTION –I

Q-1(a) State whether the following systems are Linear, Time-invariant, Causal, Stable 6

$$T(x[n]) = y[n] = (\cos \pi n) x[n].$$

(b) Give Difference equation representation of Moving-Average system. Find the frequency response of the Moving-Average system. 4

(c) What is aliasing? 1

Q-2(a) The system function of a causal linear time-invariant system is 6

$$H(z) = \frac{1 - z^{-1}}{1 + \frac{3}{4} z^{-1}},$$

The input to this system is

$$x[n] = (1/3)^n u[n] + u[-n-1].$$

(a) Find the impulse response of the system $h[n]$.

(b) Find the output $y[n]$.

(c) Is the system stable?

(b) State the properties of the Region of Convergence (R.O.C.) with suitable examples. 6

OR

Q-2(a) Determine the Z-transform and sketch the R.O.C for following sequences. 4

(i) $x[n] = (3 [2^n] - 4[3^n]) u[n].$

(ii) $x[n] = a^{-n} u[-n].$

(b) Compute the convolution using analytical method, 4
 $x[n] = \alpha^n, -3 \leq n \leq 5$, $h[n] = 1 \quad 0 \leq n \leq 4$

0 elsewhere

0 elsewhere

- (c) The frequency response of an LTI system is $H(e^{j\omega}) = e^{-j\omega/4}$, $-\pi < \omega < \pi$. Determine the output of the system, $y[n]$, when the input is $x[n] = \cos(5\pi n/2)$. 4

Q-3(a) Explain Reconstruction of Band limited signal from its samples. 4

- (b) Give decomposition of a stable, causal system into the cascade of a minimum-phase and an all-pass system; consider the two stable, causal systems specified by the system functions. 4

$$H_1(z) = \frac{(1+3z^{-1})}{(1+\frac{1}{2}z^{-1})} \quad H_2(z) = \frac{(1+\frac{3}{2}e^{j\pi/4}z^{-1})(1+\frac{3}{2}e^{-j\pi/4}z^{-1})}{(1-\frac{1}{3}z^{-1})}$$

- (c) Explain the structures for linear phase FIR system in brief. 4

OR

Q-3(a) Find the inverse Z-transform of $X[z] = \log[1+az^{-1}]$, $|z| > a$. 4

- (b) Draw the possible cascade and parallel form implementations of the LTI system with system function. 4

$$H(z) = (1+2z^{-1}+z^{-2})/(1-0.75z^{-1}+0.125z^{-2})$$

- (c) Explain polyphase decomposition in detail. 4

SECTION -II

Q-4(a) Draw and explain a radix-2 Decimation in time FFT implementation signal flow graph (butterfly diagram) for 8 point FFT. 6

- (b) Explain Least-squares optimal filtering in brief. 2

- (c) Compare FIR and IIR filter in brief. 3

Q-5(a) Two four point sequences $x[n]$ and $h[n]$ are as follows: 8

$$X[n] = \cos(\pi n/2), n = 0,1,2,3, \\ h[n] = 2^n, n = 0,1,2,3.$$

- (i) calculate the four point DFT $x[k]$ and $H[k]$
(ii) Calculate the $y[n] = x[n] \otimes h[n]$ by doing the circular convolution directly.

- (b) Explain filter design by impulse invariance method in detail. 4

OR

- Q-5(a)** Design type I chebyshev filter Given: 6
- (i) Passband ripple = 1db
 - (ii) Stop band Cutoff freq. $\Omega_s = 2000\pi$
 - (iii) Cutoff freq. passband $\Omega_p = 1000\pi$
 - (iv) Attenuation of 40dB for $\Omega \geq \Omega_s$
- (b)** Draw the butterfly diagram using composite radix DIT FFT, for $N=6$. 4
 $N=2 \times 3$ or 3×2 .
- (c)** Explain cross correlation in brief. 2
- Q-6(a)** State different applications of DSP. 4
- (b)** Given $x[n] = a^n u[n]$. 6
- (i) Determine Fourier transform $X[w]$ of $x[n]$
 - (ii) Determine discrete Fourier transform $X[k]$
 - (iii) How $X[k]$ related to $X[w]$.
- (c)** State and explain time shifting property of z-transform 2
 OR
- Q-6(a)** Draw direct form II and transposed form from the difference equations 4
 of the system
- $$w[n] = a_1 w[n-1] + a_2 w[n-2] + x[n]$$
- $$y[n] = b_0 w[n] + b_1 w[n-1] + b_2 w[n-2]$$
- (b)** Explain analysis of round off in digital filter in brief. 4
- (c)** Determine IDFT of $X[k] = \{ 2, 1+j, 0, 1-j \}$ 4